137-141/Q1-Q10, 1, 3, 9, 11, 13-16, 19, 21-24, 37, 40-42, 44, 45

§3.4 Velocity and Other Rates of Change

The instantaneous rate of change of $f(x)$ at a is the derivative

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

provided the limit exists.

* when we say "rate of change" we mean "instantaneous rate of change"

Try p 137 #2 a. $A = \pi r^2$ while $C = 2\pi r$ such that $r = \frac{C}{2\pi}$

$$\therefore A = \pi\left(\frac{C}{2\pi}\right)^2 = \pi\left(\frac{C^2}{4\pi^2}\right) = \frac{C^2}{4\pi} \text{ or } \frac{1}{4\pi}C^2$$

b. $\frac{dA}{dC} = \frac{1}{4\pi} \cdot 2C^1$

$$= \frac{C}{2\pi}$$

c. $\left.\frac{dA}{dC}\right|_{C=\pi} = \frac{\pi}{2\pi} = \frac{1}{2}$

$$\left.\frac{dA}{dC}\right|_{C=6\pi} = \frac{6\pi}{2\pi} = 3$$

d. $\frac{dA}{dC}$ would be in $\frac{in^2}{in}$

Motion Along a Line

Suppose an object is moving along a coordinate line (say an S-axis) so that we know its position S on the line as a function of time t

i.e. $S = f(t)$

The <u>displacement</u> of the object from time t to time $t + \Delta t$ is

$$\Delta S = f(t + \Delta t) - f(t)$$

The object's <u>average velocity</u> over the time interval is

$$V_{av} = \frac{\text{displacement}}{\text{travel time}} = \frac{\Delta S}{\Delta t} = \frac{f(t+\Delta t) - f(t)}{\Delta t}$$

The object's <u>instantaneous velocity</u> is

$$V(t) = \frac{ds}{dt} = \lim_{\Delta t\to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t}$$

#8 $Q(t) = 200(30-t)^2$

$$Q'(10) = \lim_{\Delta t\to 0} \frac{Q(10+\Delta t) - Q(10)}{\Delta t}$$

$$= \lim_{\Delta t\to 0} \frac{200(30-(10+\Delta t))^2 - 200(30-10)^2}{\Delta t}$$

$$= \lim_{\Delta t\to 0} \frac{200(20-\Delta t)^2 - 200(20)^2}{\Delta t}$$

$$= \lim_{\Delta t\to 0} \frac{200(400 - 40\Delta t + (\Delta t)^2) - 80000}{\Delta t}$$

$$= \lim_{\Delta t\to 0} \frac{80000 - 8000\Delta t + 200(\Delta t)^2 - 80000}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{-8000\,\Delta t + 200(\Delta t)^2}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \left(-8000 + 200\Delta t\right)$$

$$= -8000$$

∴ @ the end of 10 minutes, water is running out @ a rate of 8000 gal/min

Average rate on $[0, 10]$ $= \dfrac{Q(0+10) - Q(0)}{10}$

$$= \frac{Q(10) - Q(0)}{10}$$

$$= \frac{200(30-10)^2 - 200(30-0)^2}{10}$$

$$= \frac{200(20)^2 - 200(30)^2}{10}$$

$$= \frac{200(400) - 200(900)}{10}$$

$$= \frac{80000 - 180000}{10}$$

$$= -10000$$

∴ the average rate over the first 10 minutes is -10000
i.e. the water is running out at a rate of 10000 gal/min

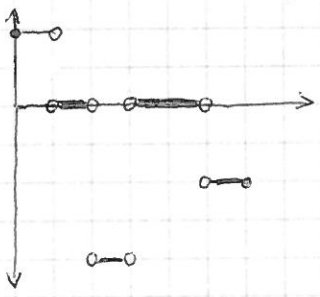
Velocity tells us two things: ① how fast an object is moving
② direction of motion

Speed, on the other hand, only tells us how fast an object is moving

$$\text{Speed} = |v(t)| = \left|\frac{ds}{dt}\right|$$

#10 a. P is moving left when the velocity (slope) is negative
$(2, 3) \cup (5, 6]$
P is moving right when the velocity (slope) is positive
$[0, 1]$
P is standing still when the velocity (slope) is zero
$(1, 2) \cup (3, 5)$

b.



<u>Acceleration</u> is the derivative of velocity w/ respect to time

$$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

Ex. A dynamite blast propels a heavy rock straight up with a launch velocity of 160 ft/sec ($\approx$ 109 mph). It reaches a height of $s = 160t - 16t^2$ ft after t seconds.

§3.4 (continued)

a. How high does the rock go?

As the rock goes straight up, it will reach its highest point when its velocity is 0.

$$v = \frac{ds}{dt} = 160 - 32t \text{ ft/sec}$$

$$160 - 32t = 0$$

$$t = 5$$

$$s_{max} = 160(5) - 16(5)^2 = 400$$

∴ the rock will reach a height of 400 ft.

b. What is the velocity and speed of the rock when it is 256 ft above the ground on the way up? On the way down?

$$256 = 160t - 16t^2$$

$$16t^2 - 160t + 256 = 0$$

$$16(t^2 - 10t + 16) = 0$$

$$16(t-8)(t-2) = 0$$

$$t = 2 \text{ and } t = 8$$

$$v|_{t=2} = 160 - 32(2) = 96 \text{ ft/sec}$$

$$v|_{t=8} = 160 - 32(8) = -96 \text{ ft/sec}$$

∴ @ both instants, the speed of the rock is 96 ft/sec

c What is the acceleration of the rock at any time t (after the blast)?

$$a = \frac{dv}{dt} = -32$$

∴ the acceleration of the rock is -32 ft/sec^2

d. When does the rock hit the ground?

Let $s = 0$

$$0 = 160t - 16t^2$$

$$0 = 16t(10 - t)$$

$$t = 0 \quad t = 10$$

∴ the rock hits the ground @ $t = 10$ sec.

p 139 #20 $s(t) = -t^3 + 7t^2 - 14t + 8$, s is in meters, t is in seconds

a. $v(t) = -3t^2 + 14t - 14$

b $a(t) = -6t + 14$

c. Let $v(t) = 0$

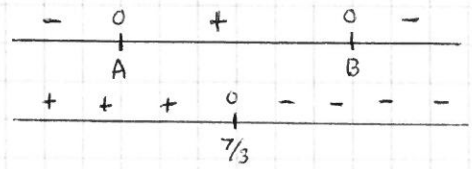
$$-3t^2 + 14t - 14 = 0$$

$$t = \frac{-14 \pm \sqrt{(14)^2 - 4(-3)(-14)}}{2(-3)}$$

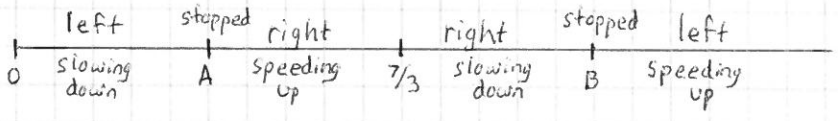
$t = 1.4514162... \rightarrow A$ and $3.21525043... \rightarrow B$

∴ the particle is at rest at $t = A$ sec and $t = B$ sec

d.



Motion of Particle



$s(0) = 8$ m

$s(A) = -0.631130...$

$s(7/3) = 0.74074...$

$s(B) = 2.11261179...$